HOMEWORK FOR EES 4800 (AY 2024/2025)

1. You are required to provide the necessary steps to get your answer.
2. Please scan/save your answers as PDF, with the file name set to be "Your name+Student ID".
3. Please submit your answers via http://inbox.weiyun.com/TptIF63G, by 11pm on 19th October 2024 (this Saturday).

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1. (1 points) Show that for any two vectors $|v_1\rangle$ and $|v_2\rangle$,

$$|\langle v_1|v_2\rangle|^2 = \langle v_1|v_2\rangle \langle v_2|v_1\rangle .$$

(Hint: represent using column vectors and row vectors with complex coefficients.)

2. (1 points) Measuring an arbitrary qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ in an arbitrary orthonormal basis $|v_0\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$ and $|v_1\rangle = \sin\theta|0\rangle - \cos\theta|1\rangle$. Expand the state $|\psi\rangle$ in the new basis, and derive the probability of obtaining the measurement outcomes. Verify that the sum of the probabilities is 1. (reminder: α and β are complex coefficients.)

3. (1 points) Consider an arbitrary orthonormal basis for qubit: $|v_0\rangle = \cos\theta|0\rangle + e^{i\phi}\sin\theta|1\rangle$. Find a proper vector $|v_1\rangle$ such that $|v_0\rangle$ and $|v_1\rangle$ constitute an orthonormal basis.

Expand the state $|\psi\rangle$ in the basis of $|v_0\rangle$ and $|v_1\rangle$, and derive the probability of obtaining the measurement outcomes. Verify that the sum of the probabilities is 1.

4. (1 points) For any $\theta \in \mathbb{R}$, consider the matrix

$$R(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix} . \tag{1}$$

Show that $R(\theta)$ is unitary.

5. (2 points) Verify the effect of Pauli X, Y, Z on the Hadamard basis states $|+\rangle$ and $|-\rangle$.

6. (4 points) We first need to introduce the concept of the Controlled-NOT or CNOT gate U_{CNOT}. This gate has two input qubits, known as the *control* qubit and the *target* qubit, respectively.

Similar to classical information processing, if the control qubit is set to 0, then the target qubit is left unchanged. If the control qubit is set to 1, then the target qubit is flipped. In equations:

$$|00\rangle \to |00\rangle ; |01\rangle \to |01\rangle ; |10\rangle \to |11\rangle ; |11\rangle \to |10\rangle . \tag{2}$$

6.1 (2 points) Give the matrix form of the CNOT gate U_{CNOT}, and prove that CNOT gate is unitary.

6.2. (2 points) Starting from the fact that $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ form a complete orthonormal basis for two-qubit systems. Prove that Bell basis states shown below also form a complete orthonormal basis for any two-qubit systems.

$$\begin{aligned}
|\phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\
|\phi^-\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\
|\psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\
|\psi^-\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)
\end{aligned} \tag{3}$$

(Hint: you may use Hadamard gate and CNOT gate for this proof.)

7. (1 points) Is $\rho = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ a valid density matrix? Why?

8. (3 points) Given a density matrix $\rho_{|\phi^+\rangle\langle\phi^+|} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$, what is the probability of obtaining $|0\rangle|0\rangle$ when we project it onto the standard basis? What is the probability of obtaining $|+\rangle|+\rangle$ when we project it onto the Hadamard basis?

9. (5 points) Let ρ be a density operator, Show that $\mathrm{Tr}(\rho^2) \le 1$, with equality if and only if ρ is a pure state.

(Hint: you may need to use the fact that density matrix ρ, a Hermitian matrix (厄米矩阵) and also a normal matrix (正规矩阵, matrix that commutes with its conjugate transpose), can be diagonalised (对角化) in an orthonormal basis via spectral decomposition (谱分解): $\rho = \sum_i p_i |i\rangle\langle i|$.)

10. (5 points) For a mixed state $\rho_{AB} = \frac{1}{6}|00\rangle\langle 00| + \frac{1}{3}|11\rangle\langle 11| + \frac{1}{2}|{+}{+}\rangle\langle{+}{+}|$, calculate the partial trace over the B subsystem.

11. (3 points) Suppose we have a qubit in the state $|0\rangle$, and we measure the observable X (Pauli matrix). What is the average value of X? What is the standard deviation of X?

12. (5 points) In class, we showed that quantum teleportation can transfer state $\alpha|0\rangle + \beta|1\rangle$ from A to B. Now let's consider Alice holds a 2-qubit system (A_0 and A_1) – an entangled state: $|\psi\rangle_{A_0A_1} = (\alpha|0\rangle_{A_0}|0\rangle_{A_1} + \beta|1\rangle_{A_0}|1\rangle_{A_1})$.

Besides, quantum system A' and B' share an EPR state: $|EPR\rangle_{A'B'} = \frac{1}{\sqrt{2}}(|0\rangle_{A'}|0\rangle_{B'} + |1\rangle_{A'}|1\rangle_{B'})$. Then Alice will perform joint measurement on A_1 and A', and project them on one of the Bell states. Alice inform Bob with the measurement result, and Bob will perform certain unitary transformation on his qubit B'.

The Bell states and unitary transforms are the same as in standard quantum teleportation example.

After these procedures, what is the quantum state shared between system A_0 and B'?

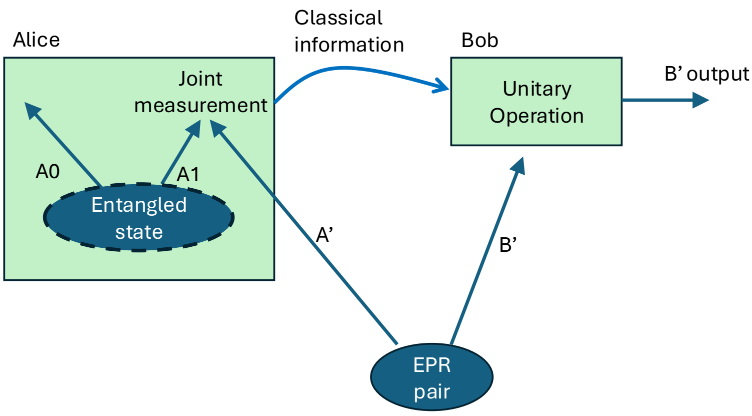

Figure 1. For question 12.

13. (8 points) In class, we performed CHSH score calculation for $|\Psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, showing that a maximum violation of $2\sqrt{2}$ is obtained.

The measurement bases for Alice in this case are $\{|0\rangle, |1\rangle\}$ (x $=$0) and $\{|+\rangle, |-\rangle\}$ (x $=$1). The measurement bases for Bob are $\{|v_1\rangle, |v_2\rangle\}$ and $\{|w_1\rangle, |w_2\rangle\}$. Here $|v_1\rangle = \cos\frac{\pi}{8}|0\rangle + \sin\frac{\pi}{8}|1\rangle$, $|v_2\rangle = -\sin\frac{\pi}{8}|0\rangle + \cos\frac{\pi}{8}|1\rangle$; $|w_1\rangle = \cos\frac{\pi}{8}|0\rangle - \sin\frac{\pi}{8}|1\rangle$, $|w_2\rangle = \sin\frac{\pi}{8}|0\rangle + \cos\frac{\pi}{8}|1\rangle$. The outcomes from Alice and Bob $a, b \in \{-1, +1\}$.

With these settings, get the CHSH score for $|\Psi'\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. (This shows maximum CHSH violation certifies both the state and measurements.)